GLOBAL SECTIONS OF THE TWISTED OBJECT ARE PSEUDO-COHERENT AND COMPUTE THE REDUCTIONS IN THE DERIVED EXISTENCE SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0DID (sources/stacks-flat/pinned-original-source.tex line 8462).

Allowed standard prerequisites (editorial).

- The object K in the derived existence situation. Same chapter, Lemma labelled lemma-compute-what-it-should-be-derived, invoked in the statement, and Situation labelled situation-existence-derived. Supplies K and the system K_n.
- Pseudo-coherence and base change of the finite level global sections. Derived Categories of Schemes, Lemma labelled perfect-lemma-flat-proper-pseudo-coherent-direct-image-general, cited at proof lines 8478-8482. Shows each M_n is pseudo-coherent over D(A_n) with compatible base change.
- Derived limits of pseudo-coherent complexes. More on Algebra, Lemma labelled more-algebra-lemma-Rlim-pseudo-coherent-gives-pseudo-coherent, cited at proof lines 8484-8488. Makes Rlim M_n pseudo-coherent and recovers its base changes.
- Commutation of global sections with derived limits. The exactness of RGamma(X, -) on D_QCoh and its commutation with products, used at proof lines 8488-8500. Identifies RGamma(X, E otimes^L K) with Rlim M_n.

Source reference. Original reference 0DIB: sources/stacks-flat/pinned-original-source.tex, source lines 8379–8379. Same-chapter pointer into the author environment situation at source line 8379; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0DIC: sources/stacks-flat/pinned-original-source.tex, source lines 8406–8406. Same-chapter pointer into the author environment lemma

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at source line 8406; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CSD: sources/stacks-chow/tags.txt, source lines 16028–16028. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8480 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CQF: sources/stacks-chow/tags.txt, source lines 15960–15960. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8486 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0DIB let K be as in Lemma 0DIC. For any perfect object E of $D(\mathcal{O}_X)$ the cohomology*

$$M = R\Gamma(X, K \otimes^{\mathbf{L}} E)$$

is a pseudo-coherent object of $D(A)$ and there is a canonical isomorphism

$$R\Gamma(X_n, K_n \otimes^{\mathbf{L}} E|_{X_n}) = M \otimes^{\mathbf{L}}_A A_n$$

in $D(A_n)$. Here $E|_{X_n}$ denotes the derived pullback of E to X_n.

Proof. Write $E_n = E|_{X_n}$ and $M_n = R\Gamma(X_n, K_n \otimes^{\mathbf{L}} E|_{X_n})$. By Derived Categories of Schemes, Lemma 0CSD we see that M_n is a pseudo-coherent object of $D(A_n)$ whose formation commutes with base change. Thus the maps $M_n \otimes^{\mathbf{L}}_{A_n} A_{n-1} \to M_{n-1}$ coming from φ_n are isomorphisms. By More on Algebra, Lemma 0CQF we find that $R\lim M_n$ is pseudo-coherent and that its base change back to A_n recovers M_n. On the other hand, the exact functor $R\Gamma(X, -) : D_{QCoh}(\mathcal{O}_X) \to D(A)$ of triangulated categories commutes with products and hence with derived limits, whence

$$R\Gamma(X, E \otimes^{\mathbf{L}} K) = R\lim R\Gamma(X, E \otimes^{\mathbf{L}} K_n) = R\lim R\Gamma(X_n, E_n \otimes^{\mathbf{L}} K_n) = R\lim M_n$$

as desired. □